

Chapter 4

Bipolar abstract argumentation systems

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1 Introduction

In most existing argumentation systems, only one kind of interaction is considered between arguments. It is the so-called attack relation. However, recent studies on argumentation [23, 34, 35, 4] have shown that another kind of interaction may exist between the arguments. Indeed, an argument can attack another argument, but it can also support another one. This suggests a notion of bipolarity, *i.e.* the existence of two independent kinds of information which have a diametrically opposed nature and which represent repellent forces.

Bipolarity has been widely studied in different domains such as knowledge and preference representation [10, 31, 25, 6]. Indeed, in [6] two kinds of preferences are distinguished: the *positive* preferences representing what the agent really wants, and the *negative* ones referring to what the agent rejects. This distinction has been supported by studies in cognitive psychology which have shown that the two kinds of preferences are completely independent and are processed separately in the mind. Another application where bipolarity is largely used is that of decision making. In [3, 19], it has been argued that when making decision, one generally takes into account some information in favour of the decisions and other pieces of information against those decisions.

In [20], a nomenclature of three types of bipolarity has been proposed using particular characteristics like *exclusivity* (can a piece of information be at the same time positive and negative), *duality* (can negative information be computed using positive information), *exhaustivity* (can information be neither positive, nor negative), computation of positive and negative information *on the same data*, computation of positive and negative information *with the same process*, *existence of a consistency constraint* between positive and negative information.

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The first type of bipolarity proposed by [20] (*symmetric univariate bipolarity*) expresses the fact that the negative feature is a reflection of the positive feature (so, they are mutually exclusive and a single bipolar univariate scale is enough for representing them).

The second one (*dual bivariate bipolarity*) expresses the fact that we need two separate scales in order to represent both features, although they stem from the same data (so, an information can be positive and negative at the same time and there is no exclusivity). However a duality must exist between both features.

And the third one (*heterogeneous bipolarity*) expresses the fact that both features do not stem from the same data though there is some minimal consistency requirement between both features.

In this chapter, we focus on the use of bipolarity in the particular domain of argumentation. In all the disparate cases, an argumentation process follows different steps: i) building the arguments and the interactions between them, ii) valuating the arguments and accounting for their interactions or not, iii) finally selecting the most acceptable arguments (or sets of arguments) and using them in order to draw a conclusion or choose a decision. Bipolarity can appear under different forms in each step of this process. In this chapter we are only concerned by the use of bipolarity at the interaction level (a more complete study of bipolarity in each step of the argumentation process is proposed in [4, 5]).

At this level, the main point is the definition of the interactions between arguments. As already said, due for instance to the presence of inconsistency in knowledge bases, arguments may be conflicting. Indeed, in all argumentation systems, an attack relation is considered in order to capture the conflicts.

However, most logical theories of argumentation assume that: if an argument a_1 attacks an argument a_3 and a_3 attacks an argument a_2 , then a_1 supports a_2 . In this case, the notion of support does not have to be formalized in a way really different from the notion of attack. It is the case of the basic argumentation framework defined by Dung, in which only one kind of interaction is explicitly represented by the *attack* relation. In this context, the support of an argument a by another argument b can be represented only if b defends a in the sense of [21]. So, support and attack are *dependent* notions. It is a parsimonious strategy, but it is not a correct description of the process of argumentation. Let us take several examples for illustrating the difference between “defence” and “support”:

Ex. 1 *We want to begin a hike. We prefer a sunny weather, then a sunny and cloudy one, then a cloudy but not rainy weather, in this order. We will cancel the hike only if the weather is rainy. But clouds could be a sign of rain. We look at the sky early in the morning. It is cloudy. The following exchange of informal arguments occurs between Tom, Ben and Dan:*

- t_1 Today we have time, we begin a hike.
- b The weather is cloudy, clouds are sign of rain, we had better cancel the hike.
- t_2 These clouds are early patches of mist, the day will be sunny, without clouds, so the weather will be not cloudy (and we can begin the hike).

- d These clouds are not early patches of mist, so the weather will be not sunny but cloudy; however these clouds will not grow, so it will not rain (and we can begin the hike).*

In this exchange, we can identify the following path of conflicts between arguments: argument *d* attacks argument *t*₂ which attacks argument *b* which in turn attacks argument *t*₁. So, with Dung's framework, argument *t*₂ is a defender of argument *t*₁, and argument *d* is a defeater of argument *t*₁. Nevertheless, arguments *t*₂ and *d* support the hike project. So, the idea of a chain of arguments and counter-arguments in which we just have to count the links and take the even one as defeaters and the odd ones as supporters is an oversimplification. So, the notion of defence proposed by [21] is not sufficient to represent support.

The following example also illustrates the need for a new kind of interaction between arguments; the following arguments are exchanged during the meeting of the editorial board of a newspaper:

Ex. 2

- a: Assuming agreement and no right of censor, information I concerning X will be published.*
b₁: X is the prime minister who may use the right of censor.
c₀: We are in democracy and even a prime minister cannot use the right of censor.
c₁: I believe that X has resigned. So, X is no longer the prime minister.
d: The resignation has been announced officially yesterday on TV Channel 1.
b₂: I is private information so X denies publication.
e: I is an important information concerning X's son.
c₂: Any information concerning a prime minister is public information.
repetition of c₁ and d: ...
c₃: But I is of national interest, so I cannot be considered as private information.

In this example, some conflicts appear: for instance, *b*₁ (resp. *b*₂) is in conflict with *a*. But we may also consider that the argument *d* given by an agent *Ag*₁ supports the argument *c*₁ given by another agent *Ag*₂. It is not only a "dialogue-like speech act": a new piece of information is really given and it is given *after* the production of the argument *c*₁. So taking *d* into account leads either to modify *c*₁, or to find a more intuitive solution for representing the interaction between *d* and *c*₁. In this case, we adopt an incremental point of view, considering that pieces of information given by different agents enable them to provide more and more arguments. We do not want to revise already advanced arguments. In contrast, we intend to represent as much as possible all the kinds of interaction between these arguments.

The last example shows how a notion of support between two arguments can be formalized with a logical representation of the structure of the arguments.

Ex. 3 *A murder has been performed and the suspects are Liz, Mary and Peter. The following pieces of information have been gathered:*

The type of murder suggests us that the killer is a female. The killer is certainly small. Liz is tall and Mary and Peter are small. The killer has long hair and uses a

lipstick. A witness claims that he saw the killer who was tall. Moreover, we are told that the witness is short-sighted, so he is no more reliable.

We use the following propositional symbols: sm (the killer is small), fem (the killer is a female), $mary$ (the killer is Mary), $lglip$ (the killer has long hair and uses a lipstick), wit (the witness is reliable), bl (the witness is short-sighted).

Here, an argument takes the form of a set of premises which entails a conclusion. So the following arguments can be formed: a_1 in favour of $mary$ (with premises $\{sm, fem, (sm \wedge fem) \rightarrow mary\}$), a_2 in favour of $\neg sm$ (with premises $\{wit, wit \rightarrow \neg sm\}$), a_3 in favour of $\neg wit$ (with premises $\{bl, bl \rightarrow \neg wit\}$), a_4 in favour of fem (with premises $\{lglip, lglip \rightarrow fem\}$).

a_3 attacks a_2 which attacks a_1 . So a_3 defends a_1 against a_2 .

Moreover, a_4 confirms the premise fem of a_1 . So, a_4 supports a_1 (in the sense that a_4 strengthens a_1). Contrastedly, a_3 defends a_1 against a_2 means that a_3 weakens the attack on a_1 brought by a_2 . So, on one side, a_1 gets a support and on the other side a_1 suffers a weakened attack.

The above examples show that the argumentation process uses arguments and counter-arguments, support and attack relations, but not always in the same way. The arguments which are available in a dynamic argumentation process rely upon premises which are not always pieces of evidence. If we accept that a new fact can undermine one of the premises (thus forming an attack), we must also accept that a new fact can enforce, or confirm a premise (thus forming a support interaction).

Following all these remarks, and in order to formalise realistic examples, a more powerful tool than the abstract argumentation framework proposed by Dung is needed. In particular, we are interested in modelling situations where two *independent* kinds of interactions are available: a positive and a negative one (see for example in the medical domain the work [23]). So, following [23, 35], we present a new argumentation framework: an abstract bipolar argumentation framework.

The chapter is organized as follows: Section 2 introduces the formal definitions of an abstract Bipolar Argumentation Framework (BAF). Then, we consider the fundamental problem of determining which arguments (or sets of arguments) can be considered as acceptable. The formal way to handle this problem is to define argumentation semantics. Section 3 introduces extension-based acceptability semantics for a BAF. These new semantics rely upon criteria which make explicitly use of both support and attack relations. In Section 4, another way to define extension-based semantics for a BAF is followed. First, a transformation of a BAF into a Dung's meta-argumentation framework is given. The support relation is used to form meta-arguments (called coalitions) in such a way that at the meta-level only conflict interactions may appear. Extensions of a BAF can then be defined from Dung's extensions of the meta framework. Section 5 addresses the question of labelling-based semantics in a BAF. Some labelling functions are proposed for a BAF. Section 6 is devoted to the related issues and to some concluding remarks. Note that the proofs of the properties given in this chapter can be found in the associated original papers.

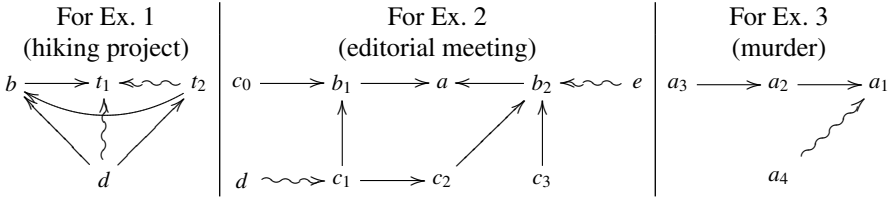
2 Abstract bipolar frameworks

An abstract bipolar argumentation framework is an extension of the basic abstract argumentation framework introduced by [21] in which a new kind of interaction between arguments is represented by the *support*¹ relation². This new relation is assumed to be totally independent of the attack relation (*i.e.* it is not defined using the attack relation). So, we have a bipolar representation of the interactions between arguments.

Def. 1 An abstract bipolar argumentation framework (BAF) $\langle \mathcal{A}, \mathcal{R}_{att}, \mathcal{R}_{sup} \rangle$ consists of: a set $\mathcal{A}$ of arguments, a binary relation $\mathcal{R}_{att}$ on $\mathcal{A}$ called the attack relation and another binary relation $\mathcal{R}_{sup}$ on $\mathcal{A}$ called the support relation. These binary relations must verify the following consistency constraint: $\mathcal{R}_{att} \cap \mathcal{R}_{sup} = \emptyset^3$. Consider a_i and $a_j \in \mathcal{A}$, $a_i \mathcal{R}_{att} a_j$ (resp. $a_i \mathcal{R}_{sup} a_j$) means that a_i attacks (resp. supports) a_j . Let $a \in \mathcal{A}$, $\mathcal{R}_{att}^-(a)$ (resp. $\mathcal{R}_{sup}^-(a)$) denotes the set of attackers (resp. supporters) of a .

In the following, we assume that $\mathcal{A}$ represents the set of arguments proposed by rational agents at a given time, so we will assume that $\mathcal{A}$ is finite.

A BAF can be represented by a directed graph $\mathcal{G}^b$ called the *bipolar interaction graph*, with two kinds of edges, one for the attack relation ($\rightarrow$) and another one for the support relation ($\rightsquigarrow$). See for instance the following representations:



In the following, we abstract from the structure of the arguments and we consider arbitrary independent relations $\mathcal{R}_{att}$ and $\mathcal{R}_{sup}$.

Def. 2 Let $\text{BAF} = \langle \mathcal{A}, \mathcal{R}_{att}, \mathcal{R}_{sup} \rangle$ be a bipolar argumentation framework and $\mathcal{G}^b$ be the associated interaction graph. Let $a, b \in \mathcal{A}$. A path from a to b in $\mathcal{G}^b$ is a sequence $(a_1, \dots, a_n)$ of elements of $\mathcal{A}$ s.t. $n \geq 2$, $a = a_1$, $b = a_n$, $a_1 \mathcal{R} a_2, \dots, a_{n-1} \mathcal{R} a_n$, with $\mathcal{R} = \mathcal{R}_{att}$ or $\mathcal{R}_{sup}$. Such a path has length $n - 1$.

Note that if $n = 2$ and $a = b$ then the path is a loop and if the relation $\mathcal{R}$ used in the loop is $\mathcal{R}_{att}$ then a is said self-attacking.

The use of bipolarity suggests new kinds of interaction between arguments: in Ex. 2, the fact that d supports an attacker of b_1 may be considered as a kind of

¹ Note that the term “support” refers to a relation between two arguments and not a relation between premises and conclusion, as in Toulmin [32].

² If the support relation is removed, we retrieve Dung’s framework.

³ In the context of the argumentation, this consistency constraint is essential: it does not seem rational to advance an argument which simultaneously attacks and supports the same other argument.

negative interaction between d and b_1 , which is however weaker than a direct attack. From a cautious point of view, such arguments cannot appear together in a same extension. In order to address this problem, a new kind of attack has been introduced [13, 14] which combines a sequence of supports with a direct attack.

Def. 3 Let $a, b \in \mathcal{A}$. There is a sequence of supports for b by a (or for short a supports b) iff there exists a sequence $(a_1, \dots, a_n)$ of elements of $\mathcal{A}$ s.t. $n \geq 2$, $a = a_1$, $b = a_n$, $a_1 \mathcal{R}_{sup} a_2, \dots, a_{n-1} \mathcal{R}_{sup} a_n$.

Def. 4 A supported attack for an argument b by an argument a is a sequence (a, x, b) of arguments of $\mathcal{A}$ s.t. a supports x^4 and $x \mathcal{R}_{att} b$.

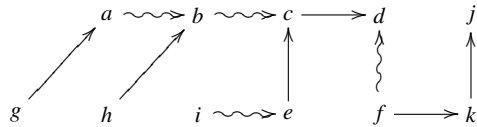
In Ex. 2, there is a supported attack for b_1 by d .

Then, taking into account attacks and sequences of supports leads to the following definitions applying to sets of arguments:

Def. 5 Let $S \subseteq \mathcal{A}$, let $a \in \mathcal{A}$. S set-attacks a iff there exists a supported attack or a direct attack for a from an element of S . S set-supports a iff there exists a sequence of supports for a from an element of S .

The above definitions are illustrated on the following example:

Ex. 4 Consider the following graph:



In this graph, the paths $a - b - c - d$ and $i - c$ correspond to supported attacks. The set $\{a, h\}$ set-attacks d and b and set-supports b and c .

3 Extension-based semantics for acceptability

In Dung's framework, the *acceptability* of an argument depends on its membership to some sets, called acceptable sets or extensions. These extensions are characterised by particular properties. It is a collective acceptability. Following Dung's methodology, we propose characteristic properties that a set of arguments must satisfy in order to be an output of the argumentation process, in a bipolar framework. We recall that such a set of arguments must be in some sense coherent and must enable to win a dispute. Maximality for set-inclusion is also often required.

Considering a BAF $\langle \mathcal{A}, \mathcal{R}_{att}, \mathcal{R}_{sup} \rangle$ and using the notion of "set-attack" and "set-support" given by Def. 5, we first investigate the notion of coherence, then we propose new semantics for acceptability in bipolar argumentation frameworks.

⁴ In the sense of Def. 3.

3.1 Managing the conflicts

In the basic argumentation framework, whatever the considered semantics, selected acceptable sets of arguments are constrained to be coherent in the sense that they must be conflict-free. In a bipolar argumentation framework, the concept of coherence can be extended along two different lines:

- forbidding not only direct attacks but also supported attacks enforces a kind of *internal* coherence: we do not accept a set S of arguments which set-attacks one of its elements (this a generalization of Dung's notion of conflict-free).
- extending the consistency constraint between support and attack relations leads to define a kind of *external* coherence: we do not accept a set S of arguments which set-attacks *and* set-supports the same argument.

Def. 6 Let $S \subseteq \mathcal{A}$. S is *+conflict-free*⁵ iff $\nexists a, b \in S$ s.t. $\{a\}$ set-attacks b .
 S is *safe*⁶ iff $\nexists b \in \mathcal{A}$ s.t. S set-attacks b and either S set-supports b , or $b \in S$.

Ex. 4 (cont'd) The set $\{h, b\}$ is not *+conflict-free* (there is a direct attack). The set $\{b, d\}$ is not *+conflict-free* since d suffers a supported attack from b . Contrastedly, $\{a, h\}$ and $\{b, f\}$ are *+conflict-free*.

The set $\{a, h\}$ is not *safe* since a supports b and h attacks b . The set $\{b, f\}$ is not *safe* since d suffers a supported attack from b and f supports d . Contrastedly, $\{g, i, h\}$ is *safe*.

Note that the notion of safe set encompasses the notion of *+conflict-free* set:

Prop. 1 ([14]) Let $S \subseteq \mathcal{A}$. If S is *safe*, then S is *+conflict-free*. If S is *+conflict-free* and closed for $\mathcal{R}_{sup}$ then S is *safe*.

Ex. 4 (cont'd) The set $\{g, h, i, e\}$ is *+conflict-free* and closed for $\mathcal{R}_{sup}$. So it is *safe*.

3.2 New acceptability semantics

According to the methodology proposed by [21], two notions play an important role in the definition of extension-based semantics: the notion of coherence, and the notion of defence (that is for short attack against attack). In a BAF, several notions of coherence, and two kinds of attack (direct and supported) are available. So several extensions of the notion of defence could be proposed. However, we have chosen to restrict to the classical defence, for the following reasons. First, the purpose of this chapter is to present some principles governing bipolar frameworks, rather than an exhaustive survey. Secondly, most of the works talking about bipolarity consider that

⁵ This notation means that checking if a set is *+conflict-free* needs to consider more conflicts than with the basic notion of conflict-free suggested by Dung.

⁶ This definition is inspired by [35] and by the notion of a controversial argument given in [21].

a support does not have the same strength as an attack. In that sense, an argument can be considered as defended if and only if its direct attackers are directly attacked.

The above remark is illustrated by the following example: $a_1 \xrightarrow{\quad} a_2 \xleftarrow{\quad} a_3$

There is a supported attack for a_1 by a_3 and no attack for a_3 . However, a_1 directly attacks a_2 and it seems sufficient to reinstate a_1 .

Let us recall the definition of defence given in [21].

Def. 7 Let $S \subseteq \mathcal{A}$. Let $a \in \mathcal{A}$. S defends a iff $\forall b \in \mathcal{A}$, if $b \mathcal{R}_{att} a$ then $\exists c \in S$ s.t. $c \mathcal{R}_{att} b$.

In the following, the concept of admissibility is first extended. The idea is to reinforce the coherence of the admissible sets. Then, extensions under the preferred semantics will be defined as maximal (for $\subseteq$) admissible sets of arguments.

Three different definitions for admissibility can be given, from the most general one to the most specific one. First, a direct translation of Dung's definition gives the definition of d-admissibility ("d" means "in the sense of Dung"). Taking into account external coherence leads to s(afe)-admissibility. Finally, external coherence can be strengthened by requiring that an admissible set is closed for $\mathcal{R}_{sup}$. So, we obtain the definition of c(losed)-admissibility.

Def. 8 Let $S \subseteq \mathcal{A}$.

S is d-admissible iff S is +conflict-free and defends all its elements.

S is s-admissible iff S is safe and defends all its elements.

S is c-admissible iff S is +conflict-free, closed for $\mathcal{R}_{sup}$ and defends all its elements.

From the above definitions, it follows that each c-admissible set is s-admissible, and each s-admissible set is d-admissible.

Def. 9 A set $S \subseteq \mathcal{A}$ is a d-preferred (resp. s-preferred, c-preferred) extension iff S is maximal for $\subseteq$ (or for short $\subseteq$ -maximal) among the d-admissible (resp. s-admissible, c-admissible) subsets of $\mathcal{A}$.

Ex. 1 (cont'd) In this case, the three semantics give the same result: $\{d, t_1\}$ is the unique d-preferred, s-preferred and c-preferred extension.

Ex. 4 (cont'd) The set $\{g, h, i, e, f, d, j\}$ is the unique c-preferred extension.

Ex. 5 Consider the BAF represented by $a \rightsquigarrow b \leftarrow h$. The set $\{a, h\}$ is the unique d-preferred extension. There are two s-preferred extensions $\{a\}$ and $\{h\}$. And there is only one c-preferred extension $\{h\}$.

One of the most important issues with regard to extensions concerns their existence. The existence of d-preferred (resp. s-preferred, c-preferred) extensions is guaranteed since the empty set is d-admissible (resp. s-admissible, c-admissible), and each d-admissible (resp. s-admissible, c-admissible) is included in a d-preferred (resp. s-preferred, c-preferred) extension. Note that analogous definitions for admissibility could be proposed using a stronger notion of defence (a stronger defence would be defined for instance by replacing *attack* with *set-attack* in Def. 7).

Considering another well-known semantics, the stable semantics, nice results can be obtained if we keep the basic definition of a stable extension, but replace *attack* with *set-attack*. It is a straightforward way to extend the stable semantics in a BAF.

Def. 10 *S is a stable extension iff S is +conflict-free and $\forall a \notin S, S$ set-attacks a.*

In the following, we restrict to acyclic BAF, in the sense that the associated interaction graph is acyclic. In Dung's basic framework, it has been proved that, in the case of an acyclic attack graph, there is always a unique stable (which is also preferred) extension. So, Def. 10 ensures the existence of a unique stable extension in an acyclic BAF⁷. However, the unique stable extension is not always safe.

Ex. 5 (cont'd) *The set $\{a, h\}$ is the unique stable extension, and it is not safe.*

Indeed, the following result can be proved:

Prop. 2 ([14]) *Let S be a stable extension. S is safe iff S is closed for $\mathcal{R}_{sup}$.*

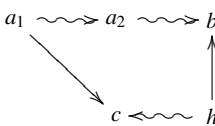
The following results enable to characterize d-preferred, s-preferred and c-preferred extensions when the BAF is acyclic:

Prop. 3 ([14]) *Let S be the unique stable extension of an acyclic BAF.*

1. *S is also the unique d-preferred extension.*
2. *The s-preferred extensions and the c-preferred extensions are subsets of S.*
3. *Each s-preferred extension which is closed for $\mathcal{R}_{sup}$ is c-preferred.*
4. *If S is safe, then S is the unique c-preferred and the unique s-preferred extension.*
5. *If A is finite, each c-preferred extension is included in a s-preferred extension.*
6. *If S is not safe, the s-preferred extensions are the subsets of S which are $\subseteq$ -maximal among the s-admissible sets.*
7. *If S is not safe, and A is finite, there is only one c-preferred extension.*

Ex. 5 (cont'd) *$\{h\}$ is the only s-preferred extension which is also closed for $\mathcal{R}_{sup}$. So, $\{h\}$ is the unique c-preferred extension.*

Ex. 6 *Consider the BAF represented by:*



$\{a_1, a_2, h\}$ is the only d-preferred extension. $\{a_1, a_2\}$ and $\{h\}$ are the only two s-preferred extensions. None of them is closed for $\mathcal{R}_{sup}$. $\emptyset$ is the unique c-preferred extension. If we add an isolated argument a_3 (for which no interaction exists with the other available arguments), then we obtain: $\{a_1, a_2, a_3, h\}$ is the only d-preferred extension. $\{a_1, a_2, a_3\}$ and $\{h, a_3\}$ are the only two s-preferred extensions. None of them is closed, and $\{a_3\}$ is the unique c-preferred extension.

⁷ We instantiate Dung's AF with the relation set-attacks and the resulting graph is still acyclic.

The above discussion enables to draw the following conclusions. In the particular case of an acyclic BAF, two semantics present nice features: the stable semantics and the c-preferred semantics. If we are interested in internal coherence only, we will have to determine the unique stable extension, which is also the unique d-preferred extension. If we are interested in a more constrained concept of coherence, we will compute the unique c-preferred extension.

4 Turning a bipolar framework into a Dung meta-framework

The extension-based acceptability semantics introduced in Section 3 rely upon criteria which make explicitly use of support and attack relations, through the concept of supported attack. Here, we follow another way to define extension-based semantics for a BAF. First, a transformation of a BAF into a Dung's meta-argumentation framework is given. This meta-argumentation framework consists only of a set of meta-arguments (called coalitions), and a conflict relation between these meta-arguments. The attack relation of the initial BAF will appear only at the meta-level. As a consequence, a meta-argument will gather arguments which are not in conflict. The support relation of the initial BAF will not appear at the meta-level, but will be used to gather arguments in a coalition. The idea is that a meta-argument makes sense only if its members are somehow related by the support relation. So, the two fundamental principles governing the definition of a coalition are: the *Coherence principle* (there is no direct attack between two arguments of a same coalition) and the *Support principle* (if two arguments belong to a same coalition, they must be somehow, directly or indirectly, related by the support relation).

4.1 The concept of coalition

Consider $\text{BAF} = \langle \mathcal{A}, \mathcal{R}_{\text{att}}, \mathcal{R}_{\text{sup}} \rangle$ represented by the graph $\mathcal{G}^b$. $\mathcal{G}_{\text{sup}}^b$ will denote the partial graph representing the partial system $\langle \mathcal{A}, \mathcal{R}_{\text{sup}} \rangle$ ⁸. AF will denote the partial argumentation system $\langle \mathcal{A}, \mathcal{R}_{\text{att}} \rangle$ associated with BAF and represented by the partial graph denoted by $\mathcal{G}_{\text{att}}^b$.

Def. 11 $C \subseteq \mathcal{A}$ is a coalition of BAF iff: (i) The subgraph of $\mathcal{G}_{\text{sup}}^b$ induced by C is connected; (ii) C is conflict-free⁹ for AF; (iii) C is $\subseteq$ -maximal among the sets satisfying (i) and (ii).

⁸ We consider that the reader knows the basic concepts of graph theory (chain, connexity, ...). See for instance [7] for a background on graph theory.

⁹ In the basic sense proposed by Dung.

Note that when $\mathcal{R}_{\text{att}}$ is empty, the coalitions are exactly the connected components¹⁰ of the partial graph $\mathcal{G}_{\text{sup}}^b$.

Prop. 4 ([17]) *An argument which is not self-attacking is in at least one coalition.*

Ex. 7 Consider the BAF:

The coalitions are: $C_1 = \{b, c, d\}$, $C_2 = \{i\}$, $C_3 = \{a, b\}$, $C_4 = \{e, f, g, h\}$

The following result shows that coalitions can be restated in terms of connected components of an appropriate subgraph. By the way, it gives a constructive way for computing coalitions.

Prop. 5 ([17]) $C \subseteq \mathcal{A}$ is a coalition of BAF iff: (i) There exists $S \subseteq \mathcal{A}$ $\subseteq$ -maximal conflict-free for AF s.t. C is a connected component of the subgraph of $\mathcal{G}_{\text{sup}}^b$ induced by S and (ii) C is $\subseteq$ -maximal among the subsets of $\mathcal{A}$ satisfying (i).

Prop. 5 suggests a procedure for computing the coalitions of BAF:

- Step 1: Consider AF and determine the maximal conflict-free sets for AF.
- Step 2: For each set of arguments S_i obtained at Step 1, determine the connected components of the subgraph of $\mathcal{G}_{\text{sup}}^b$ induced by S_i .
- Step 3: Keep the $\subseteq$ -maximal sets obtained at Step 2.

The notion of conflict-free set is related to the notion of independent set:

Prop. 6 ([17]) Let $S \subseteq \mathcal{A}$. S is conflict-free for AF iff S is an independent subset of $\mathcal{A}$ in the graph $\mathcal{G}_{\text{att}}^b$.

So, S is $\subseteq$ -maximal conflict-free for AF iff S is a $\subseteq$ -maximal independent set of vertices in the graph $\mathcal{G}_{\text{att}}^b$ and Step 1 of the computational procedure consists in determining all the $\subseteq$ -maximal independent subsets of $\mathcal{G}_{\text{att}}^b$. Remark that the time complexity of the best algorithms providing all the $\subseteq$ -maximal independent sets is exponential. Note also that there exist several algorithms in the literature for finding all the $\subseteq$ -maximal independent sets (see for instance the work of J.M. Nielsen [26]). We also know that:

- For Step 2, a depth-first exploration of a graph provides the connected components in linear time $\mathcal{O}(\text{number of vertices} + \text{number of edges})$.
- And for Step 3, maximization with respect to $\subseteq$ is also an exponential process.

¹⁰ Let $G = (V, E)$ be a graph. Let $S \subseteq V$. S is a *connected component* of G iff the subgraph of G induced by S is connected and there exists no $S' \subseteq V$ s.t. $S \subset S'$ and the subgraph of G induced by S' is connected.

4.2 A meta-argumentation framework

Let $C(\mathcal{A})$ denote the set of coalitions of BAF. We define a conflict relation on $C(\mathcal{A})$ as follows.

Def. 12 Let C_1 and C_2 be two coalitions of BAF. C_1 C-attacks C_2 iff there exists an argument a_1 in C_1 and an argument a_2 in C_2 s.t. $a_1 \mathcal{R}_{att} a_2$.

It can be proved that:

Prop. 7 ([17]) Let C_1 and C_2 be two distinct coalitions of BAF. If $C_1 \cap C_2 \neq \emptyset$ then C_1 C-attacks C_2 or C_2 C-attacks C_1 .

So a new argumentation framework $\text{CAF} = \langle C(\mathcal{A}), \text{C-attacks} \rangle$ can be defined, referred to as the coalition framework associated with BAF.

Ex. 7 (cont'd) In this example, CAF can be represented by (by abusing notations, $\rightarrow$ represents the attack relation in BAF and also the C-attack relation in CAF):

$$C_3 \longleftarrow C_1 \longrightarrow C_4 \longleftarrow C_2$$

Dung's definitions apply to CAF, and it can be proved that:

Prop. 8 ([17]) Let $\{C_1, \dots, C_p\}$ be a finite set of distinct coalitions. $\{C_1, \dots, C_p\}$ is conflict-free for CAF iff $C_1 \cup \dots \cup C_p$ is conflict-free for AF.

So, CAF is a “meta-argumentation” framework with a set of “meta-arguments” (the set of coalitions $C(\mathcal{A})$) and a “meta-attack” relation on these coalitions (the C-attacks relation). A coalition gathers arguments which are close in some sense and can be produced together. However, as coalitions may conflict, following Dung's methodology, preferred and stable extensions of CAF can be computed. Such extensions will contain coalitions which are collectively acceptable. The last step consists in gathering the elements of the coalitions of an extension of CAF. By this way, the best groups of arguments (w.r.t. the given interaction relations) will be selected.

Def. 13 Let $S \subseteq \mathcal{A}$. S is a Cp-extension (Cp means “Coalition-preferred”) of BAF iff there exists $\{C_1, \dots, C_p\}$ a preferred extension of CAF s.t. $S = C_1 \cup \dots \cup C_p$. S is a Cs-extension (Cs means “Coalition-stable”) of BAF iff there exists $\{C_1, \dots, C_p\}$ a stable extension of CAF s.t. $S = C_1 \cup \dots \cup C_p$.

When the only preferred extension of CAF is the empty set, we define the empty set as the unique Cp-extension of BAF.

Ex. 7 (cont'd) There is only one preferred extension of CAF, which is also stable: $\{C_1, C_2\}$. So, $S = \{b, c, d, i\}$ is the Cp-extension (and also the Cs-extension) of BAF.

Some nice properties of Dung's basic framework are preserved:

- A BAF has always a (at least one) Cp-extension. It is a consequence of Def. 13.
- In contrast, there does not always exist a Cs-extension of BAF. The reason is that there may be no stable extension of CAF.
- Each Cs-extension is also a Cp-extension. The converse is false.

- There cannot exist two Cp-extensions s.t. one strictly contains the other one. It follows from Def. 11 and 13.

However, other properties are lost. A Cp-extension is not always admissible for AF, and a Cs-extension is not always a stable extension of AF:

Ex. 8 Consider the BAF represented by: $a \longrightarrow b \leftarrow \sim c \rightsquigarrow d \longrightarrow e$

The coalitions are: $C_1 = \{a\}$, $C_2 = \{b, c, d\}$, $C_3 = \{e\}$. And the associated CAF can be represented by: $C_1 \longrightarrow C_2 \longrightarrow C_3$

There is only one preferred extension of CAF, which is also stable: $\{C_1, C_3\}$. So, $S = \{a, e\}$ is the Cp-extension (and also the Cs-extension) of BAF. We have $d \mathcal{R}_{att} e$, but a does not defend e against d (neither by a direct attack, nor by a supported attack, though a attacks an element of the coalition which attacks e). So, S is not admissible for AF. S does not contain c , but there is no attack (no supported attack) of an element of S against c . So, S is not a stable extension of AF.

Note that a coalition is considered as a whole and its members cannot be used separately in an attack process. Ex. 8 suggests that admissibility is lost due to the size of the coalition $\{b, c, d\}$, and that it would be more fruitful to consider two independent coalitions $\{c, b\}$ and $\{c, d\}$. A new formalization of coalitions in terms of conflict-free maximal support paths has been proposed in [17]. However, it does not enable to recover Dung's properties.

Note that the lost of admissibility in Dung's sense is not surprising: admissibility is lost because it takes into account "individual" attack and defence, whereas, with meta-argumentation and coalitions, "collective" attack and defence are considered.

5 Labellings in bipolar frameworks

This section addresses the question of labelling-based semantics in a BAF. A labelling-based semantics relies upon a set of labels and is defined by specifying the criteria for assigning labels to arguments. An example of labelling-based semantics in a basic argumentation framework is given by [22] with the robust semantics. More generally, several approaches have been proposed for valuing the arguments in a classical argumentation framework (for example [24, 29, 30, 8, 15, 1, 33]). In some of them, the value of an argument depends on its interactions with the other arguments; in other ones, it depends on an intrinsic strength of the argument. Besides, Karacapilidis & Papadias [23] have proposed an argumentation web-tool, named HERMES, for decision making in the medical field. Taking into account notions of attacks and supports between arguments, this system permits the expression and the weighting of arguments. The basic elements of this system are: a *solution* (an answer to the question which is discussed) and a *position* (expressing the support for, or the opposition to a solution or to another position). HERMES can label the solutions and the positions by the status "active" or "inactive". At the end of the discussion, the "active" (resp. "inactive") solutions are accepted (resp. rejected). An

“active” solution is a recommended choice among the other solutions concerning a same question. Different recursive labellings are proposed in HERMES. But, the value of a position p depends only on the *active* positions which are linked to p in the acyclic discussion graph, and the value of a position is always binary.

In this section, we propose a limited use of the notion of labelling-based semantics for a BAF: we show how bipolar interactions can be used for defining valuations over the set of arguments, *i.e.* functions which assign a value to each argument of the BAF (a further step would be to use such a valuation in order to select arguments, that is to completely define labelling-based semantics in a BAF, in an analogous way as what has been done in [16] for basic argumentation frameworks).

The approach presented here (see [13]) has the following features: the valuation process takes place before the selection process; the valuation process makes use of a rich set of values and not only two as in HERMES (so, it is called a gradual valuation); the value assigned to an argument takes into account all the direct attackers and supporters of this argument (it is not the case in HERMES in which the value of an argument only depends on the *active* positions); so it is called a local valuation.

This proposition extends the works [22, 8, 16] to bipolar argumentation frameworks as defined in Section 2. It follows the same principles as those already described in [15] augmented with new principles corresponding to the “support” information. So, the three underlying principles for a gradual interaction-based local valuation are:

- **P1:** The value of an argument depends on the values of its direct attackers and of its direct supporters.
- **P2:** If the quality of the support (resp. attack) increases then the value of the argument increases (resp. decreases).
- **P3:** If the quantity of the supports (resp. attacks) increases then the quality of the support (resp. attack) increases.

The value of an argument is obtained with the composition of several functions:

- one for aggregating the values of all the direct attackers; this function computes the value of the “attack”
- one for aggregating the values of all the direct supporters; this function computes the value of the “support”
- one for computing the effect of the attack and of the support on the value of the argument.

In the respect of the previous principles, we assume that there exists a completely ordered set $\mathcal{V}$ with a minimum element $\mathcal{V}_{\min}$ and a maximum element $\mathcal{V}_{\max}$. The following formal definition for a gradual local valuation can be given.

Def. 14 Let $\langle \mathcal{A}, \mathcal{R}_{att}, \mathcal{R}_{sup} \rangle$ be a bipolar argumentation framework. Let $a \in \mathcal{A}$ with $\mathcal{R}_{att}^-(a) = \{b_1, \dots, b_n\}$ and $\mathcal{R}_{sup}^-(a) = \{c_1, \dots, c_p\}$.

A local gradual valuation on $\langle \mathcal{A}, \mathcal{R}_{att}, \mathcal{R}_{sup} \rangle$ is a function $v : \mathcal{A} \rightarrow \mathcal{V}$ s.t.:

$$v(a) = g(h_{sup}(v(c_1), \dots, v(c_p)), h_{att}(v(b_1), \dots, v(b_n))) \text{ with}$$

the function h_{att} (resp. h_{sup}): $\mathcal{V}^* \rightarrow \mathcal{H}_{att}$ (resp. $\mathcal{V}^* \rightarrow \mathcal{H}_{sup}$)¹¹ valuing the quality of the attack (resp. support) on a , and the function g : $\mathcal{H}_{sup} \times \mathcal{H}_{att} \rightarrow \mathcal{V}$ with $g(x, y)$ increasing on x and decreasing on y . The function h , $h = h_{att}$ or h_{sup} , must satisfy:

1. if $x_i \geq x'_i$ then $h(x_1, \dots, x_i, \dots, x_n) \geq h(x_1, \dots, x'_i, \dots, x_n)$,
2. $h(x_1, \dots, x_n, x_{n+1}) \geq h(x_1, \dots, x_n)$,
3. $h() = \alpha \leq h(x_1, \dots, x_n) \leq \beta$, for all $x_1, \dots, x_n$ ¹².

Def. 14 produces a generic local gradual valuation. Let us give two instances of the generic definition, to illustrate the different principles.

- A first instance is defined by $\mathcal{H}_{att} = \mathcal{H}_{sup} = \mathcal{V} = [-1, 1]$ interval of the real line, $h_{att}(x_1, \dots, x_n) = h_{sup}(x_1, \dots, x_n) = \max(x_1, \dots, x_n)$, and $g(x, y) = \frac{x-y}{2}$ (so, we have $\alpha = -1$, $\beta = 1$ and $g(\alpha, \alpha) = 0$).
- Another one is defined by $\mathcal{V} = [-1, 1]$ interval of the real line, $\mathcal{H}_{att} = \mathcal{H}_{sup} = [0, \infty[$ interval of the real line, $h_{att}(x_1, \dots, x_n) = h_{sup}(x_1, \dots, x_n) = \sum_{i=1}^n \frac{x_i+1}{2}$, and $g(x, y) = \frac{1}{1+y} - \frac{1}{1+x}$ (so, we have $\alpha = 0$, $\beta = \infty$ and $g(\alpha, \alpha) = 0$).

The following table shows the values computed with both instances on some simple examples:

Example	with 1 st instance	with 2 nd instance
No attack, no support: a	$v(a) = 0$	$v(a) = 0$
Direct attack: $a \longrightarrow b$	$v(b) = -0.5$	$v(b) = -0.33$
Direct support: $a \rightsquigarrow b$	$v(b) = 0.5$	$v(b) = 0.33$
Defence: $a \longrightarrow b \longrightarrow c$	$v(c) = -0.25$	$v(c) = -0.25$
Sequence of supports: $a \rightsquigarrow b \rightsquigarrow c$	$v(c) = 0.75$	$v(c) = 0.4$
Supported attack: $a \rightsquigarrow b \longrightarrow c$	$v(c) = -0.75$	$v(c) = -0.4$

Ex. 1 (cont'd) With the first (resp. second) instance, $v(t_1) = \frac{1}{4}$ (resp. $\frac{37}{154}$).

Ex. 5 (cont'd) With the first and the second instances, $v(b) = 0$. In this case, there is a perfect equilibrium¹³ between support and attack.

A local gradual valuation defined as above satisfies the following properties [13]:

- If $\mathcal{R}_{att}^-(a) = \mathcal{R}_{sup}^-(a) = \emptyset$ then $v(a) = g(\alpha, \alpha)$.
- If $\mathcal{R}_{att}^-(a) \neq \emptyset$ and $\mathcal{R}_{sup}^-(a) = \emptyset$ then $v(a) = g(\alpha, y) \leq g(\alpha, \alpha)$ for $y \geq \alpha$.
- If $\mathcal{R}_{att}^-(a) = \emptyset$ and $\mathcal{R}_{sup}^-(a) \neq \emptyset$ then $v(a) = g(x, \alpha) \geq g(\alpha, \alpha)$ for $x \geq \alpha$.

¹¹ $\mathcal{V}^*$ denotes the set of the finite sequences of elements of $\mathcal{V}$, including the empty sequence. $\mathcal{H}_{att}$ and $\mathcal{H}_{sup}$ are ordered sets.

¹² So, α is the minimal value for an attack (resp. a support) – i.e. there is no attack (resp. no support) –, and β is the maximal value for an attack (resp. a support).

¹³ Note that it is not necessarily the case, and an appropriate choice of the function g enables to give more importance to the attack than to the support.

And we have the following comparative scale¹⁴:

$$\mathcal{V}_{\min} \leq \underset{(\text{for } y \geq \alpha)}{g(\alpha, y)} \leq g(\alpha, \alpha) \leq \underset{(\text{for } x \geq \alpha)}{g(x, \alpha)} \leq \mathcal{V}_{\max}$$

Moreover the valuation proposed in Def. 14 satisfies the principles **P1** to **P3** (see [13] for a more detailed discussion).

6 Related issues and conclusion

In this chapter, an extension of [21]’s abstract argumentation framework has been proposed in order to take into account two kinds of interaction between arguments modelled with a support relation and an attack relation. In this abstract BAF, two issues have been considered:

- taking into account bipolarity for defining acceptability semantics: either by enforcing the coherence of the admissible sets, or by turning a BAF into a meta-argumentation framework using the concept of coalition;
- taking into account bipolar interactions for proposing gradual labellings for the arguments.

6.1 Related issues about acceptability and bipolarity

Deflog [35]: DEFLOG argumentation system enables to express a support or an attack between sentences in the language, with a new sentence using specific connectors (one for each kind of interaction). Examples of sentences (with $\rightarrow$ for the attack relation and $\leadsto$ for the support relation) are: $a, b, (a \leadsto b), (a \rightarrow b), (c \leadsto (a \leadsto b)), (d \rightarrow (a \leadsto b))$. In DEFLOG, the notions of sequence of supports and of supported attacks can be retrieved but at the language level (between sentences). Moreover, the notion of conflict-free set proposed in DEFLOG corresponds to the notion of safe set (no sentence which is, at the same time, supported and attacked by the set).

DEFLOG enables to define the dialectical interpretations (or extensions) of a given set of sentences S : an extension is built from a partition (J, D) of S such that J is conflict-free and attacks the sentences of D .

Note that the attack relation and the support relation are explicitly expressed in the sentences. So, one can have an extension of a set S s.t. some supported sentences by J do not belong to S . DEFLOG extensions correspond to [21]’s stable extensions for DEFLOG theories that do not go beyond the expressiveness of Dung’s argumentation frameworks, and note that a Dung’s AF can always be expressed in DEFLOG. So in this precise sense, DEFLOG’s extensions are a faithful generalization of Dung’s stable extensions, allowing more expressiveness. Moreover, [35] gives also a faithful generalization of Dung’s preferred extensions.

¹⁴ Using this scale, the values $\leq$ (resp. $\geq$) to $g(\alpha, \alpha)$ are considered as negative (resp. positive) ones even if $g(\alpha, \alpha) \neq 0$.

Evidence-based argumentation [28]: In this work, the fundamental claim is that an argument cannot be accepted unless it is supported by evidence. So, special arguments are distinguished: the *prima-facie* arguments (which do not require any support to stand).

Arguments may be acceptable only if they are supported (indirectly) by *prima-facie* arguments. This is evidential support. Moreover, only supported arguments may attack other arguments.

Then, the notion of defence is rather complex: A set of arguments S defends an argument a if S provides evidential support for a and S invalidates each attack on a (either by a direct attack on the attacker of a or by rendering this attack unsupported).

Following our definitions, a BAF is an abstract framework, where arguments may stand and attack with or without support. However, evidential reasoning as proposed by [28] could also be handled in a BAF in the following way: Given X a set of arguments (which are considered as *prima-facie* arguments in a given application), a notion of evidential support can be defined via a sequence of supports from an argument of X . Then, the notion of attack can be restricted so that attackers be elements of X , or receive evidential support from X . Finally, instead of choosing the classical definition for “ S defends a ” (as presented in Def. 7), it can be required first that S provides support for a and secondly that for each supported attack on a , one argument of the sequence of supports is directly attacked by S .

6.2 Related issues about coalitions of arguments

Another way for defining acceptability semantics in a bipolar framework is to turn a *bipolar* argumentation framework into a *meta-argumentation framework*. This transformation has the following characteristics: the support relation is used in order to identify “coalitions” (sets of arguments which can be used together without conflict and which are related by the support relation) and the attack relation is used in order to identify conflicts between coalitions and then to define new acceptability semantics as in Dung’s framework.

The concept of coalition has already been related to argumentation.

Collective argumentation framework [9, 27]: A collective argumentation framework is an abstract framework where the initial data are a set of arguments and a binary “attack” relation between *sets* of arguments. The key idea is the following: a set of arguments can produce an attack against other arguments, which is not reducible to attacks between particular arguments. That is in agreement with our notion of coalition, since in our work, a coalition is considered as a whole and its members cannot be used separately in an attack process. The proposal by Nielsen and Parsons is similar to Bochman’s proposal. Both proposals take the attacks between sets of arguments as initial data, and define semantics by properties on subsets of arguments. However, Nielsen and Parsons propose an abstract framework which allows sets of arguments to attack single arguments only, and they stick as close

as possible to the semantics provided by Dung. In contrast, Bochman departs from Dung's methodology and give new specific definitions for stable and admissible sets of arguments. Our proposal essentially differs from collective argumentation in two points. First, we keep exactly Dung's construction for defining semantics, but we apply this construction in a meta-argumentation framework (the coalition framework). The second main difference lies in the meaning of a coalition: we intend to gather as many arguments as possible in a coalition, and a coalition cannot be broken in the defence process.

Generation of coalition structures in MAS [18, 2]: In multi-agent systems (MAS), the coalition formation is a process in which independent and autonomous agents come together to act as a collective. A coalition structure (CS) is a partition of the set of agents into coalitions. Each coalition has a value (the utility that the agents in the coalition can jointly get minus the cost which this coalition induces for each agent). So the value of a CS is obtained by aggregating the values of the different coalitions in the structure. One of the main problems is to generate a preferred CS, that is a structure which maximizes the global value. Recently, [2] has proposed an abstract system where the initial data are a set of coalitions equipped with a conflict relation. A preferred CS is a subset of coalitions which is conflict-free and defends itself against attacks. Coalitions may conflict for instance if they are non-disjoint or if they achieve a same task.

However, the generation of the coalitions is not studied in [2]. So, one perspective is to apply our work to the formation of coalitions taking into account interactions between the agents. Arguments represent agents in that case. Indeed, it is very important to put together agents which want to cooperate ("supports" relation) and to avoid gathering agents who do not want to cooperate ("attacks" relation). Then, the concept of Cp-extension provides a tool for selecting the best groups of agents (w.r.t. the given interaction relations).

More generally, the work reported here is generic and takes place in abstract frameworks, since no assumption is made on the nature of the arguments. Arguments may have a logical structure such as a pair ⟨explanation, conclusion⟩, may just be positions advanced in a discussion, or may be agents interacting in a multi-agent system. All that we need is the bipolar interaction graph describing how the arguments under consideration are interrelated. We think that this generic work should stimulate discussion across boundaries.

6.3 Related issues about valuation and bipolarity

Most works about valuations of arguments take place in the basic framework. Some of them consider intrinsic valuations, which express to what extent an argument increases the confidence in the statement it promotes. Other approaches consider interaction-based valuations. These approaches usually differ in the set of values which are available.

However, very few works have been interested in valuations which handle both support and attack interactions. Most of these works have been developed for specific applications.

Medical applications: The most influential work has been proposed in HERMES system [23]. But there is no graduality (only two possible values with HERMES), and some parts of the interacting arguments are not taken into account for the computation of the value. See in Section 5.

Valued maps of argumentations: The bipolar valuation in argumentation has been used for a collective annotation of documents.

Collective annotation models supporting exchange through discussion threads. A discussion thread is initiated by an annotation about a given document. Then, users can reply with annotations which confirm or refute the previous ones. Annotations are associated with a social validation which provides a synthetic view of the discussions. The purpose of this validation is to identify annotations which are globally confirmed by the discussion thread. It can also take into account an intrinsic value of the annotations.

In [11, 12] a discussion thread is modelled by a BAF. The set of arguments contains the nodes of the thread. Pairs of the support (resp. attack) relation correspond to replies in the thread of the confirm (resp. refute) type. The social validation of a given annotation is computed with the local bipolar valuation. Moreover, the bipolar valuation procedure has been slightly modified in order to take into account an intrinsic value of each annotation.

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